UCK 364E Design of Machine Elements

Lecture 2: Fatigue — the Stress-Life and Strain-Life Approaches

Cyclic loading, S–N curves, endurance limit, notch and surface effects, mean stress, cumulative damage, ε–N

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Learning objectives

By the end of this lecture you should be able to

1. describe the three stages of fatigue damage and explain why fatigue is a *local*, surface-initiated phenomenon;
2. define the quantities that describe a cycle (σ_a, σ_m, R, A) and convert between them;
3. construct an S–N (Wöhler) curve for a component from material data, applying the Marin modifying factors for surface, size, loading, temperature and reliability;
4. compute the fatigue notch factor K_f from K_t using notch sensitivity, and explain why $K_f < K_t$;
5. apply the Goodman, Gerber, Soderberg and Morrow mean-stress corrections and say which one aerospace practice uses;
6. accumulate damage over a variable-amplitude spectrum with Palmgren–Miner, and count cycles with the rainflow method;
7. explain the strain-life (Coffin–Manson) approach, Neuber's rule and when ε–N is required instead of S–N;
8. relate all of the above to the safe-life philosophy of CS-25.571 and to the scatter factors used in certification.

1 Why fatigue governs aircraft structure

In Lecture 1 every static margin we computed was large: the fuselage skin had an ultimate margin of 2.9, the actuator rod end 5.6. Nothing in an aircraft is sized by a single application of load. What sizes it is the *repetition* of loads far below the static allowable: one pressurisation cycle per flight, a gust spectrum through every hour of cruise, a landing impact per flight, vibration from the propeller or rotor at hundreds of hertz.

The central fact

A metal loaded repeatedly to a stress well below its yield strength will eventually crack. There is no "safe" static margin that protects against this; the only protections are to keep the cyclic stress low enough, to limit the number of cycles, or to find the crack before it becomes critical.

Fatigue was the lesson of the de Havilland Comet (1954) and it remains the dominant design driver for pressurised fuselages, wing lower surfaces, lugs and fittings, and every joint. Two historical approaches to fatigue life exist and this lecture covers both:

- the **stress-life (S–N)** approach, valid when stresses stay elastic and lives are long ($N > 10^3$–10^4 cycles), which is the workhorse for most machine elements;
- the **strain-life (ε–N)** approach, needed where local plasticity occurs at notches and lives are short, and which is the physically correct description of crack initiation.

The third approach, **fracture mechanics** (crack *growth* rather than initiation), is the subject of the next lecture.

1.1 The three stages of fatigue

1. **Crack initiation**. Cyclic slip in favourably oriented surface grains produces intrusions and extrusions; a microcrack of the order of one grain diameter forms, almost always at a free surface and almost always at a stress concentration. In a well-finished laboratory specimen this stage can consume 90% of the life; in a notched, corroded or poorly finished part, almost none of it.
2. **Stable crack growth**. The crack grows a small increment per cycle, normal to the maximum principal stress, leaving *striations* — one per cycle — and, on a macroscopic scale, *beach marks* that record changes in load level. This stage is described quantitatively by fracture mechanics.
3. **Final fracture**. The remaining section can no longer carry the peak load and fails statically, in one cycle, leaving a rough, fibrous or crystalline area.

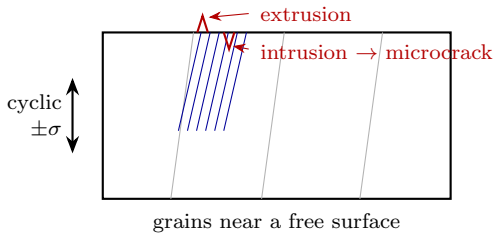


Figure 1: Crack nucleation. Cyclic slip in favourably oriented surface grains is irreversible: each cycle leaves a step, the steps pile up into an extrusion and its matching intrusion, and the intrusion is a notch one grain deep that the following cycles turn into a microcrack.

A fatigue fracture surface is therefore readable: a smooth, beach-marked region (growth) adjoining a rough region (final fracture), with the initiation site at the focus of the beach marks. Reading a fracture surface is the first step in every failure investigation.

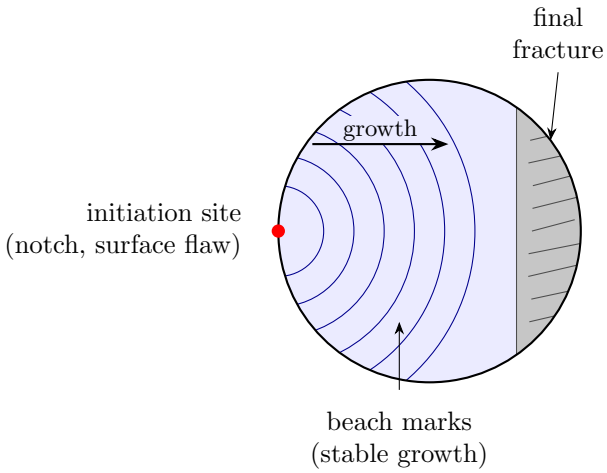


Figure 2: A fatigue fracture surface is readable: beach marks centred on the initiation site record stable growth, the rough remainder is the final static fracture, and the relative areas indicate how highly the part was loaded.

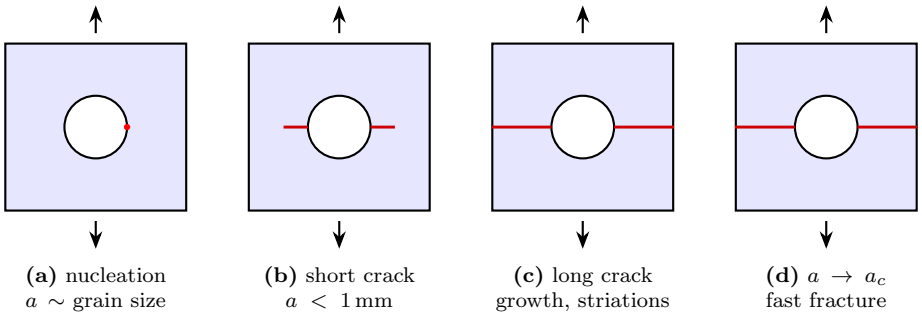


Figure 3: The life of a crack at a fastener hole. (a) and (b) are initiation, which the S–N and ε–N methods of this lecture predict; (c) and (d) are crack growth, the subject of the next lecture. Nothing visible happens until late in the life.

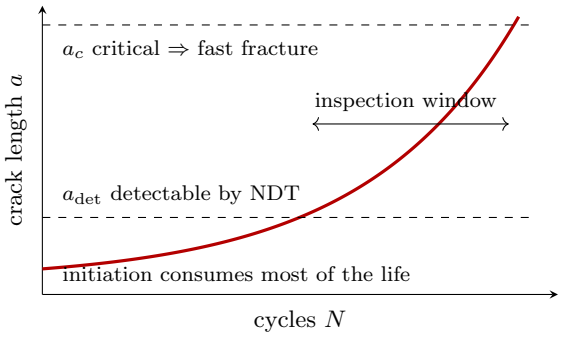


Figure 4: Crack length against cycles. Growth accelerates, so most of the life is spent at a size nobody can detect. The usable inspection window runs from the detectable size a_{det} to the critical size a_c, and the inspection interval is set so that at least two inspections fall inside it.

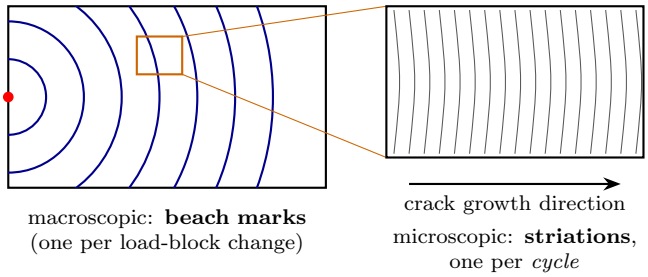


Figure 5: Two scales on the same fracture surface. Beach marks are visible to the eye and record changes of load *block* (a flight, a shutdown); striations need an electron microscope and record individual *cycles*. Striation spacing lets an investigator reconstruct how long the crack had been growing.

2 Describing a cycle

A constant-amplitude cycle between $\sigma_{\min}$ and $\sigma_{\max}$ is described by

$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}, \qquad \sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2}, \qquad \Delta\sigma = \sigma_{\max} - \sigma_{\min} = 2\sigma_a, \tag{1}$$

and by the dimensionless ratios

$$R = \frac{\sigma_{\min}}{\sigma_{\max}}, \qquad A = \frac{\sigma_a}{\sigma_m} = \frac{1-R}{1+R}. \tag{2}$$

Common cases: $R = -1$ fully reversed (the reference condition for material data, $\sigma_m = 0$); $R = 0$ zero-to-tension (typical of a pressurisation cycle or a landing load); $R = 0.1$ the usual test condition for aerospace coupons; $R = 1$ static. Tensile mean stress is damaging, compressive mean stress is beneficial — a fact exploited deliberately by shot peening and hole cold-expansion.

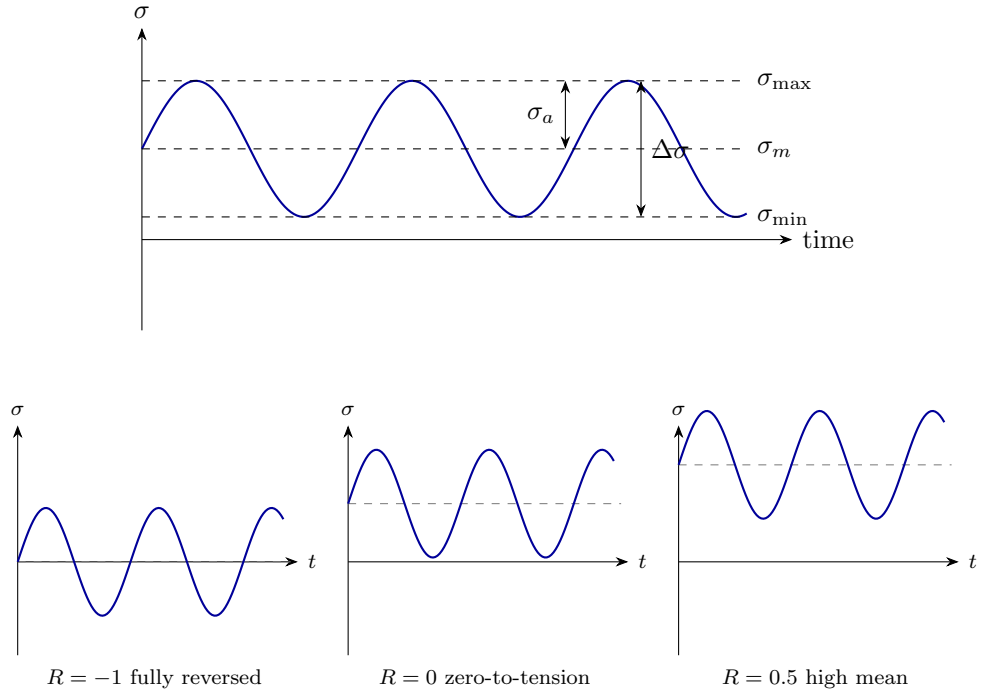


Figure 6: The same stress amplitude at three mean stresses. All three have identical σ_a and very different lives: moving from $R = -1$ to $R = 0.5$ at constant amplitude can cost an order of magnitude in cycles. The mean-stress corrections of §3.5 exist to account for this.

3 The stress-life approach

3.1 The S–N curve

Rotating-bending or axial tests on polished specimens at $R = -1$ give the Wöhler or S–N curve: alternating stress S_a against cycles to failure N, both on log axes. Three regions matter:

- **Low-cycle fatigue**, $N < 10^3$: plasticity dominates, S–N is unreliable, use ε–N.
- **High-cycle fatigue**, $10^3 < N < 10^6$–10^7: approximately a straight line on log–log axes.
- **The long-life region**: ferrous alloys and titanium show a true **endurance limit** S'_e at 10^6–10^7 cycles below which life is effectively infinite. **Aluminium alloys do not.** Their curve keeps falling, so an aluminium "fatigue strength" must always be quoted at a stated life, conventionally 5×10^8 cycles.

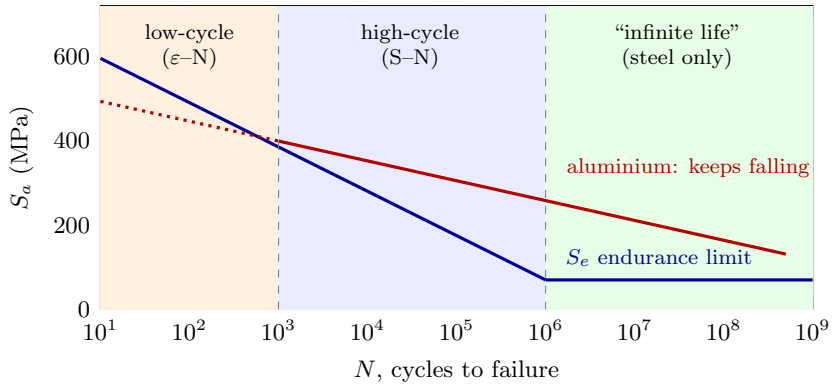


Figure 7: The S–N curve and its three regions. Below 10^3 cycles plasticity dominates and the strain-life method is required; between 10^3 and the knee the curve is a straight line on log–log axes; beyond the knee steel and titanium have a true endurance limit and aluminium does not. This is why aircraft aluminium structure is designed to a finite life with inspections rather than to "infinite life".

For a first estimate when no data exist, the classical correlations are

$$S'_e \approx \begin{cases} 0.5\,S_{ut} & \text{steel, } S_{ut} \le 1400\,\text{MPa} \\ 700\,\text{MPa} & \text{steel, } S_{ut} > 1400\,\text{MPa} \\ 0.4\,S_{ut} & \text{cast iron, copper alloys} \\ S_f@5\times10^8 \approx 0.4\,S_{ut} & \text{wrought aluminium (a fatigue } \textit{strength}\text{, not a limit)} \end{cases} \tag{3}$$

and, for the high-cycle line, $S_{1000} \approx 0.9\,S_{ut}$ for bending ($0.75\,S_{ut}$ axial). Writing the line as $S_a = aN^b$ between these two anchors gives

$$b = -\frac{1}{3}\log_{10}\left(\frac{S_{1000}}{S_e}\right), \qquad a = \frac{S_{1000}}{(10^3)^b}, \qquad N = \left(\frac{S_a}{a}\right)^{1/b}. \tag{4}$$

Use correlations only as a sanity check

For certification you use measured data: MMPDS Chapter 9 gives S–N curves for every alloy and product form at several K_t and R values, with the equivalent-stress parameter $S_{eq} = S_{\max}(1-R)^p$ fitted to the data. A correlation from S_{ut} is a first-week estimate, not an allowable.

3.2 Modifying factors: from specimen to component

The laboratory specimen is polished, 7.6 mm in diameter, loaded in rotating bending at room temperature, with 50% survival probability. A real part is none of these. The Marin approach

multiplies the specimen value by factors:

$$S_e = k_a\, k_b\, k_c\, k_d\, k_e\, k_f\, S'_e. \tag{5}$$

k_a **surface**
$k_a = a\, S_{ut}^{b}$ with $(a, b) = (1.58, -0.085)$ ground, $(4.51, -0.265)$ machined or cold-drawn, $(57.7, -0.718)$ hot-rolled, $(272, -0.995)$ as-forged, for S_{ut} in MPa. The factor is brutal: an as-forged surface on a 1200 MPa steel gives $k_a \approx 0.23$. Fatigue starts at the surface, so the surface is where the designer's money goes: grinding, polishing, shot peening (which adds compressive residual stress and can give $k_a > 1$ in effect), and avoiding machining marks transverse to the load.

k_b **size** $k_b = 1.24\, d^{-0.107}$ for $2.79 \le d \le 51$ mm, $k_b = 1.51\, d^{-0.157}$ for larger; $k_b = 1$ for axial loading. Larger parts contain more material at high stress and therefore more potential initiation sites. For non-round sections use the equivalent diameter from the 95%-stressed area.

k_c **loading**
1 for bending, 0.85 for axial, 0.59 for pure torsion (when using a uniaxial S'_e).

k_d **temperature**
≈ 1 up to about 150 °C for aluminium and 450 °C for steel; below that, strength rises slightly but toughness falls.

k_e **reliability**
1.000, 0.897, 0.868, 0.814, 0.753, 0.702 for 50, 90, 95, 99, 99.9, 99.99% survival, assuming an 8% standard deviation of endurance limit.

k_f **miscellaneous**
corrosion (which removes the endurance limit entirely — a corroded steel has no safe stress), fretting (3–10× life reduction), plating (cadmium and chromium plating reduce fatigue strength by 10–35%), residual stress, and frequency effects.

Aerospace practice

Airframe stress offices rarely run the Marin chain. They use MMPDS S–N curves measured on specimens with the relevant K_t, apply a knock-down for surface finish and environment from company allowables, and then apply the *scatter factor* of the certification method (typically 3 to 5 on life for safe-life structure, with full-scale test evidence). The Marin factors remain the right mental model for *why* the knock-downs exist, and they are the method you need for non-airframe machine elements: shafts, springs, gear teeth, actuator rods.

3.3 Notches: K_t, K_f and notch sensitivity

In Lecture 1 we met K_t, the elastic stress concentration factor, and noted that static ductile design usually ignores it. Fatigue cannot: the local stress at the notch root is what drives initiation. But the full K_t is not realised either, because the stress gradient at a sharp notch means only a small volume sees the peak. The **fatigue notch factor** is

$$K_f = 1 + q\,(K_t - 1), \qquad 0 \le q \le 1, \tag{6}$$

where q is the **notch sensitivity**. Neuber's expression is

$$q = \frac{1}{1 + \sqrt{a}/\sqrt{r}}, \tag{7}$$

with r the notch root radius and $\sqrt{a}$ the Neuber material constant ($\sqrt{a} \approx 0.25$–$0.5\,\sqrt{\text{mm}}$ for high-strength steels, larger for soft steels, $\approx 0.6\,\sqrt{\text{mm}}$ for aluminium). The implications are worth stating explicitly:

- Sharp notches in *soft* materials are relatively forgiving (q small); the same notch in a high-strength alloy is not ($q \to 1$). **The stronger the material, the more it is punished by a notch.** This is why a 1800 MPa 4340 part with a sharp fillet can be worse in fatigue than a 900 MPa part with a generous one.
- K_f is applied to the *alternating* stress. Common practice (and the conservative one) is to apply it to the mean stress as well unless local yielding at the notch is accounted for.
- Equivalently one may divide the endurance limit by K_f: $S_e^{\text{notched}} = S_e/K_f$. The two bookkeeping conventions must not both be used at once.

3.4 Mean stress

Material data come at $R = -1$; real cycles have a mean. The classical corrections relate the alternating stress σ_a and mean stress σ_m at failure:

$$\text{Goodman (modified):} \quad \frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{n} \tag{8}$$

$$\text{Gerber:} \quad \frac{n\,\sigma_a}{S_e} + \left(\frac{n\,\sigma_m}{S_{ut}}\right)^2 = 1 \tag{9}$$

$$\text{Soderberg:} \quad \frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_y} = \frac{1}{n} \tag{10}$$

$$\text{Morrow:} \quad \frac{\sigma_a}{S_e} + \frac{\sigma_m}{\sigma'_f} = \frac{1}{n} \tag{11}$$

Goodman is the straight line from S_e on the σ_a axis to S_{ut} on the σ_m axis: simple, conservative, and the one used almost universally in design. Gerber (a parabola) fits ductile-steel data better and is less conservative. Soderberg uses S_y and additionally guarantees no yielding, which makes it the most conservative. Morrow, with the true fracture strength σ'_f, agrees best with steel data and is the natural partner of the strain-life method. For compressive mean stress ($\sigma_m < 0$) all of these are taken as $\sigma_a = S_e$: compression does not reduce fatigue strength and may improve it, but it is normal practice not to take credit for the improvement.

A first-cycle yield check must accompany the fatigue check: the Langer line $\sigma_a + \sigma_m = S_y$ truncates the diagram near the σ_m axis.

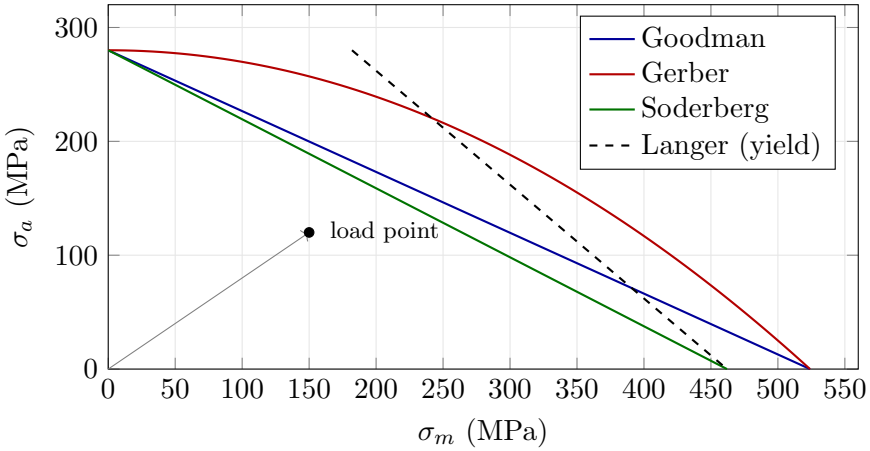


Figure 8: Constant-life diagram for 7075-T6 ($S_{ut} = 524$, $S_y = 462$, $S_e = 280$ MPa assumed). The safety factor is the ratio in which the load line OP is extended to meet the chosen criterion.

3.5 Cumulative damage and cycle counting

Real spectra are not constant amplitude. The **Palmgren–Miner** rule assumes each cycle consumes a fraction of life independent of order:

$$D = \sum_i \frac{n_i}{N_i}, \qquad \text{failure at } D = 1. \tag{12}$$

Miner's rule is linear, ignores sequence effects and ignores cycles below the endurance limit — all three assumptions are known to be wrong. Test data scatter between $D = 0.3$ and $D = 3$ at failure, and a high-low sequence is more damaging than low-high because the large cycles initiate a crack that the small ones then grow. Practice compensates with the scatter factor, and the Haibach modification keeps a shallower slope below the knee instead of truncating. Despite everything, Miner remains the standard tool.

To use Eq. (12) on a measured stress history you must first reduce the history to a set of closed cycles. The accepted algorithm is **rainflow counting** (ASTM E1049), which pairs each local peak with a valley such that each closed hysteresis loop in the material corresponds to one counted cycle; this is the only counting method consistent with the material's stress–strain memory. Level-crossing and peak counting are older, cruder alternatives.

For aircraft, the spectrum is built from the mission profile: ground–air–ground (GAG) cycle, one per flight, usually the largest single contributor; cabin pressurisation, one per flight; gust and manoeuvre loads from statistical models (the classical exceedance curves); taxi, landing impact and braking loads; and local vibration. A *flight-by-flight* spectrum ordered as the aircraft will actually experience it is the standard input to both analysis and the full-scale fatigue test.

4 Worked examples

Example 2.1 – Endurance limit of a machined steel shaft

A rotating shaft of 4340 steel, $S_{ut} = 1240\,\text{MPa}$, $S_y = 1100\,\text{MPa}$, diameter 30 mm, machined finish, operating at room temperature, is required to have 99% reliability in rotating bending.
Specimen limit: $S'_e = 0.5 \times 1240 = 620\,\text{MPa}$.
Surface: $k_a = 4.51 \times 1240^{-0.265} = 4.51 \times 0.1529 = 0.690$.
Size: $k_b = 1.24 \times 30^{-0.107} = 1.24 \times 0.6968 = 0.864$.
Loading: $k_c = 1$ (bending). Temperature: $k_d = 1$. Reliability: $k_e = 0.814$ (99%).
$S_e = 0.690 \times 0.864 \times 1 \times 1 \times 0.814 \times 620 = 297\,\text{MPa}$.
The chain of factors has cut the endurance limit to 48% of the specimen value — before any notch is considered. Add a shoulder fillet with $K_t = 1.8$ and $q = 0.85$: $K_f = 1 + 0.85(0.8) = 1.68$, so the usable alternating stress at the fillet is $297/1.68 = 177\,\text{MPa}$, i.e. 14% of the tensile strength. Choosing a stronger steel would raise S_{ut} but barely raise this number, because k_a falls and q rises with strength. Improving the fillet radius and the surface finish would do far more.

Example 2.2 – Fuselage skin under pressurisation cycling

Return to the fuselage of Lecture 1: 2024-T3, $r = 2.0\,\text{m}$, $t = 1.6\,\text{mm}$, $\Delta p = 60\,\text{kPa}$, hoop stress 75 MPa once per flight from zero (so $\sigma_a = \sigma_m = 37.5\,\text{MPa}$, $R = 0$). At a riveted lap joint take $K_t = 3.0$ for the hole with $q \approx 0.9$, giving $K_f = 2.8$.
Local alternating stress: $K_f \sigma_a = 2.8 \times 37.5 = 105\,\text{MPa}$; local mean $= 105\,\text{MPa}$.
For 2024-T3, $S_{ut} = 434\,\text{MPa}$ and the smooth fatigue strength at 10^6 cycles is about 140 MPa at $R = -1$. Goodman-correcting to $\sigma_m = 105\,\text{MPa}$:

$$\sigma_{a,\text{allow}} = 140\left(1 - \frac{105}{434}\right) = 106\,\text{MPa}.$$

The applied local alternating stress 105 MPa sits essentially *on* the 10^6-cycle line, i.e. the joint has a nominal life of order 10^6 flights with no margin for scatter, and a design service goal of 60 000 flights requires a scatter factor of about 17 on life. That is why real lap joints need either lower hoop stress, cold-expanded holes, bonded doublers, or — the route actually taken — an explicit damage-tolerance justification with inspections (next lecture). Note also what this example hides: the lap joint carries secondary bending from the eccentric load path, which can double the local stress. Lecture 1's error gallery showed this as the "stiffness jump" and "bolt in bending" cases.

Example 2.3 – Miner's rule on a simple spectrum

A fitting in 7075-T6 experiences, per flight: 1 GAG cycle at $\sigma_a = 120$ MPa, 10 manoeuvre cycles at 80 MPa, and 200 gust cycles at 40 MPa, all at $R = 0$ and already corrected for mean stress and K_f. The component S–N line is $S_a = 1100\,N^{-0.13}$ MPa.
Lives: $N = (S_a/1100)^{1/(-0.13)}$.
$N_1 = (120/1100)^{-7.69} = 2.5 \times 10^7$; $N_2 = (80/1100)^{-7.69} = 5.7 \times 10^8$; $N_3 = (40/1100)^{-7.69} = 1.2 \times 10^{11}$.
Damage per flight: $D = 1/2.5{\times}10^7 + 10/5.7{\times}10^8 + 200/1.2{\times}10^{11} = 4.0\times10^{-8} + 1.8\times10^{-8} + 1.7\times10^{-9} = 5.9\times10^{-8}$.
Flights to $D = 1$: 1.7×10^7. With a scatter factor of 4, the certified life would be 4.2×10^6 flights.
Observe the structure of the answer: the single large cycle does 67% of the damage, the 200 small ones do 3%. This is the general pattern and it has a practical consequence — effort spent reducing the peak loads pays far more than effort spent on the small ones, and a spectrum truncated at the low end changes the answer very little (but truncating the *high* end is a serious error).

5 The strain-life approach

5.1 Why S–N is not enough

At a notch root under a large load, the local material yields even though the nominal stress is elastic. Stress then no longer characterises the loading — the notch root is strain-controlled by the surrounding elastic material, and two different notches with the same local stress can have very different local strain. For $N < 10^4$ cycles, for notches in ductile alloys, and for any thermal- or displacement-controlled loading, the strain-life approach is the correct one. It is also the modern basis of crack-*initiation* prediction in commercial fatigue codes.

5.2 Cyclic stress–strain and the Coffin–Manson equation

Metals do not retain their monotonic stress–strain curve under cycling: they cyclically harden or soften until a stable hysteresis loop forms. The stable loop is described by the *cyclic* stress–strain curve

$$\frac{\Delta\varepsilon}{2} = \frac{\Delta\sigma}{2E} + \left(\frac{\Delta\sigma}{2K'}\right)^{1/n'}, \tag{13}$$

with cyclic strength coefficient K' and cyclic strain-hardening exponent n' ($n' \approx 0.1$–0.2 for most metals). Life is then given by the sum of an elastic and a plastic term, the **Coffin–Manson–Basquin** equation:

$$\frac{\Delta\varepsilon}{2} = \underbrace{\frac{\sigma'_f}{E}(2N_f)^b}_{\text{elastic, Basquin}} + \underbrace{\varepsilon'_f(2N_f)^c}_{\text{plastic, Coffin–Manson}}, \tag{14}$$

where σ'_f is the fatigue strength coefficient (close to the true fracture stress), b the fatigue strength exponent (≈ -0.05 to -0.12), ε'_f the fatigue ductility coefficient (close to the true fracture strain) and c the fatigue ductility exponent (≈ -0.5 to -0.7). $2N_f$ is the number of load *reversals*.

The two terms cross at the *transition life* N_t, typically 10^3–10^4 cycles. Below it, plasticity dominates and *ductility* controls life; above it, elasticity dominates and *strength* controls life. This single plot explains a recurring design dilemma: for a low-cycle application (a landing-gear part, a thermally cycled engine component) a tough, ductile alloy outlives a strong, brittle one, while for a high-cycle application the reverse is true.

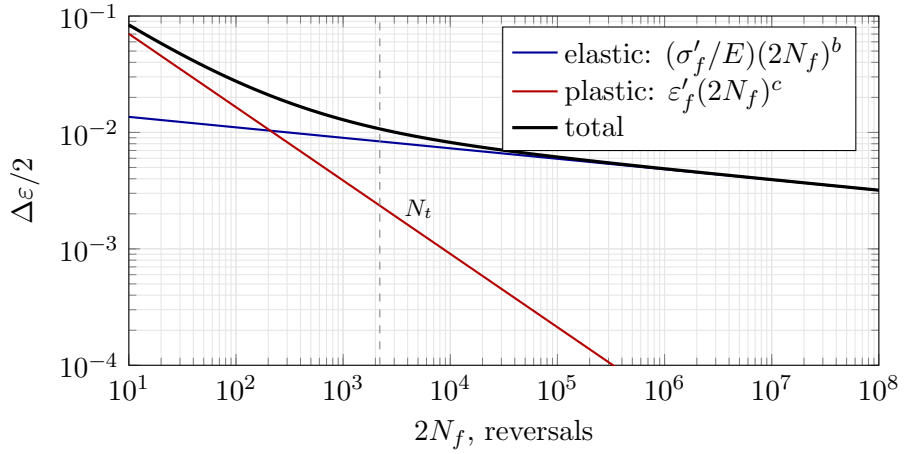


Figure 9: Strain-life curve, schematic for an aluminium alloy. Left of the transition life ductility governs; right of it, strength governs.

5.3 Neuber's rule

To use Eq. (14) for a notched part we need the local stress and strain from the nominal ones. **Neuber's rule** states that the geometric mean of the stress and strain concentration factors equals K_t (or K_f in fatigue):

$$K_f^2 = K_\sigma K_\varepsilon \quad \Longrightarrow \quad \sigma\,\varepsilon = \frac{(K_f\,S)^2}{E}, \tag{15}$$

where S is the nominal stress. Solving this hyperbola simultaneously with the cyclic stress–strain curve gives the local (σ, ε); the strain amplitude then enters Eq. (14). Glinka's energy-density rule is the less conservative alternative. Mean-stress effects enter through the Morrow correction, which replaces σ'_f with $(\sigma'_f - \sigma_m)$, or through the Smith–Watson–Topper parameter $\sigma_{\max}\varepsilon_a$.

Example 2.4 – Neuber at a notch

A 7075-T6 plate ($E = 71.7\,\text{GPa}$, $K' = 670\,\text{MPa}$, $n' = 0.07$, $\sigma'_f = 1200\,\text{MPa}$, $b = -0.09$, $\varepsilon'_f = 0.3$, $c = -0.63$) has a hole with $K_f = 2.6$ and is cycled with nominal $S_a = 150\,\text{MPa}$ (29% of S_{ut}).
Neuber: $\sigma\varepsilon = (2.6 \times 150)^2/71\,700 = 2.12\,\text{MPa}$.
Cyclic curve: $\varepsilon = \sigma/71\,700 + (\sigma/670)^{1/0.07}$. Solving the pair iteratively gives $\sigma \approx 379\,\text{MPa}$, $\varepsilon \approx 5.59 \times 10^{-3}$; the purely elastic prediction would have been $\sigma = K_f S = 390\,\text{MPa}$ and $\varepsilon = 5.44 \times 10^{-3}$, so the notch root has yielded and the strain is 3% higher than elastic theory says.
Life from Eq. (14) with $\Delta\varepsilon/2 = 5.59 \times 10^{-3}$: solving $5.59 \times 10^{-3} = 0.01674(2N_f)^{-0.09} + 0.3(2N_f)^{-0.63}$ gives $2N_f \approx 2.5 \times 10^5$ reversals, i.e. $N_f \approx 1.2 \times 10^5$ cycles to initiation. Note that the plastic term contributes only 2% of the strain amplitude here: at 10^5 cycles we are well to the right of the transition life, which is why the S–N approach would also have

worked.
Two remarks. First, the notch root yields on the first cycle even though the nominal stress is less than a third of S_{ut}; this is the normal condition at an aircraft fastener hole, and it is why the nominal stress alone tells you nothing about fatigue. Second, this is a life to *initiation* of a small crack, not to failure — the remaining life is a crack-growth problem.

6 Fatigue and the certification codes

CS-25.571 (and FAR 25.571) require an evaluation of the strength, detail design and fabrication of the structure, to show that catastrophic failure due to fatigue will be avoided throughout the operational life. Three philosophies are recognised:

- **Safe-life**: the part is retired at a life demonstrated by analysis and test, divided by a scatter factor (commonly 3–5 on a full-scale test result, more for analysis alone). No cracks are assumed to exist. Used where inspection is impractical: landing-gear structure, some rotorcraft components, engine discs.
- **Fail-safe**: a second load path carries limit load after the failure of any single element. The lug example from Lecture 1's error gallery is exactly this.
- **Damage tolerance**: cracks are assumed to exist from manufacture; inspections are scheduled so that a crack is found before it reaches critical size. This is the default for transport-category metallic structure and the subject of the next lecture.

The scatter factor exists precisely because of everything in this lecture that was approximate: the scatter in S–N data, the non-linearity of damage accumulation, the uncertainty in K_f, surface and environment effects, and the variability of the usage spectrum.

Design levers in order of effectiveness

1. Remove the stress concentration: larger radii, gentler transitions, no sharp changes of section.
2. Reduce the *peak* cycles in the spectrum (they dominate the damage sum).
3. Improve the surface: better finish, shot peening, cold-expanded holes, interference-fit fasteners (a cold-expanded hole can give a 3–5× life improvement, the largest single gain available to the designer).
4. Introduce compressive residual stress and avoid tensile mean stress.
5. Protect from corrosion and fretting; specify the plating and sealing.
6. Only then: change the material or add thickness.

7 Problems for self-study

1. A 20 mm ground shaft of 4130 steel ($S_{ut} = 670$ MPa) rotates in bending. Compute S_e for 95% reliability. Then find the allowable bending moment for infinite life at a fillet with $K_t = 2.1$, $r = 2$ mm.
2. A 7075-T6 link carries a fluctuating axial load between 10 kN and 60 kN, with $K_f = 1.9$ and cross-section 200 mm^2. Using $S_e = 150$ MPa (already modified), find the Goodman and Gerber safety factors and the Langer yield factor.
3. For the spectrum in Example 2.3, recompute the life if the GAG cycle is reduced to 100 MPa by a design change. By what factor does the life improve? Now instead delete all 200 gust cycles and comment.
4. Write a short Python function that, given $(\sigma'_f, b, \varepsilon'_f, c, E)$ and a strain amplitude, returns N_f by solving Eq. (14) numerically. Verify it against Example 2.4, then plot the elastic, plastic and total curves and find the transition life for 2024-T3 and for 4340.

5. Implement rainflow counting (ASTM E1049) for a short stress history of your choice and compare the damage sum against simple peak counting. Which is more conservative, and why?
6. Explain, in terms of the Marin factors and notch sensitivity, why a highly polished 2000 MPa steel bolt with a rolled thread can outperform a 1200 MPa bolt with a cut thread by a factor of three in fatigue, while the same two bolts differ by less than a factor of two in static strength.

Reading

Shigley (Budynas & Nisbett) Ch. 6 (fatigue failure resulting from variable loading), §§6.1–6.17. Dowling, *Mechanical Behavior of Materials*, Ch. 9–10 (S–N) and 14 (strain-life). Norton Ch. 6. MMPDS-2023 Ch. 9 (fatigue data and the equivalent-stress parameter). Peterson (Pilkey) for K_t. ASTM E466 (axial fatigue testing), E606 (strain-controlled testing), E739 (statistical analysis of S–N data), E1049 (cycle counting). EASA CS-25.571 and AMC 25.571 in the Easy Access Rules for CS-25; FAA AC 25.571-1D.